

INNOVA JUNIOR COLLEGE
JC2 PRELIMINARY EXAMINATION
in preparation for General Certificate of Education Advanced Level
Higher 1

CANDIDATE
NAME

Suggested Answer

CLASS

INDEX NUMBER

CHEMISTRY

8873/02

XX XXX 2018

Paper 2 Structured Questions

2 hour

Candidates answer on the question paper.

Additional Materials: Data Booklet

READ THESE INSTRUCTIONS FIRST

Write your index number, name and civics group.
Write in dark blue or black pen.
You may use pencil for any diagrams, graphs or rough working.
Do not use staples, paper clips, highlighters, glue or correction fluid.

Section A

Answer **all** questions in the space provided.

Section B

Answer **1 out of 2** questions on writing paper provided..

A *Data Booklet* is provided.

You are advised to show all working in calculations.
You are reminded of the need for good English and clear presentation in your answers.
You are reminded of the need for good handwriting.
Your final answers should be in 3 significant figures.

You may use a calculator.

The number of marks is given in brackets [] at the end of each question or part question.

Paper 2	67
Paper 1	33
Overall	100
Overall Grade	

For Examiner's Use	
Section A	
1	12
2	12
3	10
4	12
5	14
Section B (Please circle)	
6 or 7	20
Significant figures and units	
Handwriting	
P2 Total	80
P1 Total	30

This document consists of **12** printed pages.



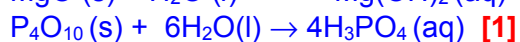
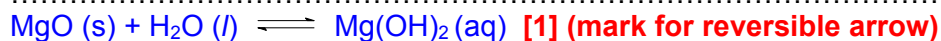
Section A

Answer all the questions.

- 1 Magnesium, the eighth most abundant element on Earth, has diverse applications such as in building or medicine.

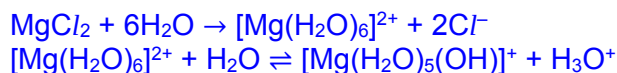
- (a) (i) Write equations for the reactions of the oxides of magnesium and phosphorus with water.

.....[2]

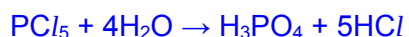


- (ii) Write equations for the reactions of magnesium chloride and phosphorus chloride with water respectively. Include in your answer, the pH of the resultant solutions.

.....[4]



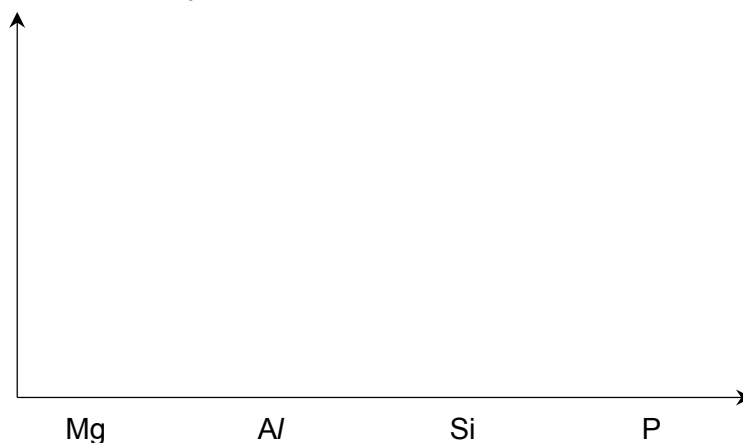
Magnesium chloride solution: **pH 6.5.**



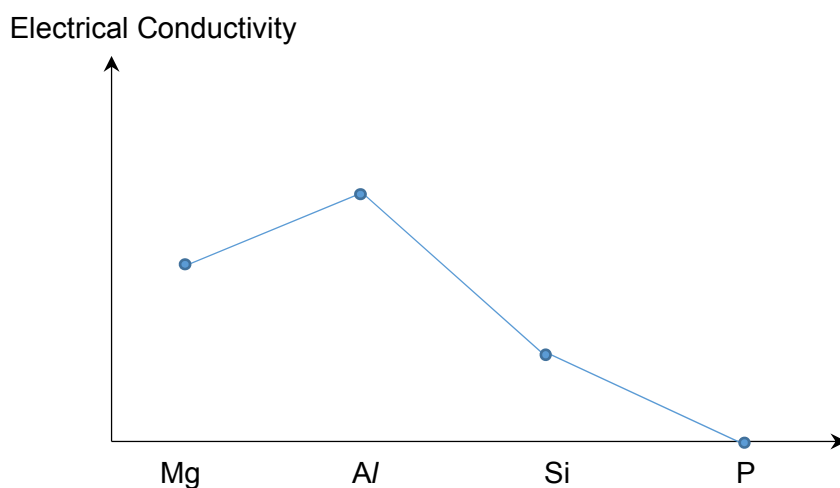
Phosphorus chloride solution: **pH 1-2.**

- (ii) On the axes below, sketch the electrical conductivity of the period 3 elements from magnesium to phosphorous.

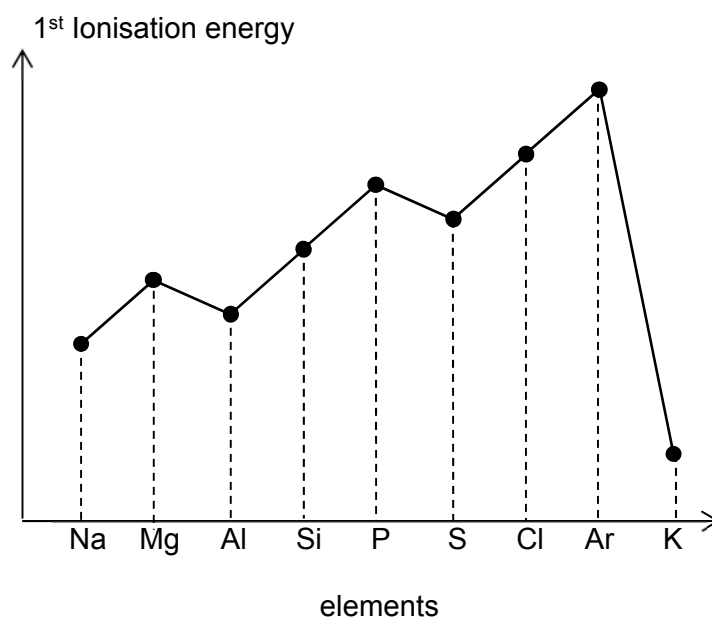
Electrical Conductivity



[1]



- (b) The following graph shows the trend of the first ionisation energies of the elements from sodium to potassium.



- (i) Explain the difference between the values of their first ionisation energies for **each** of the pairs of elements listed below. You should use a different explanation for each pair.

- sodium and potassium

.....

.....

.....

.....

.....[2]



K has one more quantum shell of electrons than Na, valence electron of K is further away from the nucleus and less strongly attracted to the nucleus. Less energy is required to remove the valence electrons from K.

Hence 1st IE of K is lower than that of Na.

- magnesium and aluminium

.....

.....[1]



The 3p electron of Al is further away from the nucleus and has higher energy than the 3s electron of Mg. Less energy is required to remove the 3p electron from Al. Thus, 1st IE of Mg is higher than that of Al.

- (ii) When a beam of $_{20}\text{Ca}^{2+}$ particles travels through a uniform electric field which is at right angles to its direction of travel, it is deflected at an angle of $+10.0^\circ$.

Determine the angle of deflection of a beam of $_{8}\text{O}^{2-}$ particles if it travels at the same speed through the same electric field.

[2]
 [Total: 12]

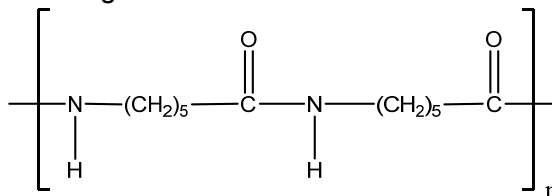
$$\text{angle of deflection} \propto \frac{q}{m}$$

Particle	z/m ratio	angle of deflection
Ca^{2+}	+2/40.1	$+10.0^\circ$
O^{2-}	-2/16	$[(-2/16 / (+2/40))] \times (+10.0) = \underline{-25^\circ}$

correct magnitude
correct sign

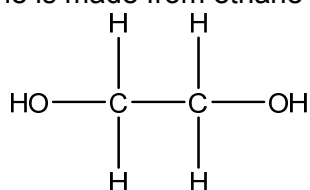
- 2 Synthetic polymer fibres are man-made, formed entirely by chemical synthesis. Demand for synthetic fibres has increased over the last few decades as they are durable, less expensive and more stain resistant than natural fibres. Terylene and Nylon 6 are examples of polymeric synthetic fibre.

- Nylon 6 has the following structure.

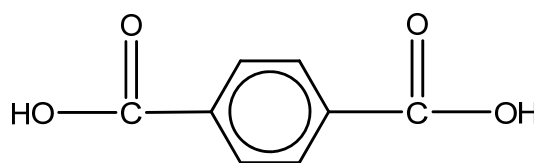


Nylon 6

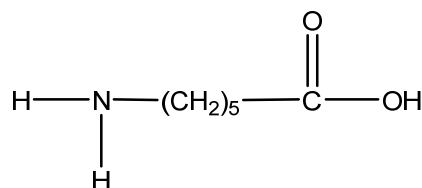
- Terylene is made from ethane-1,2-diol and benzene-1,4-dicarboxylic acid.



and



- (a) Draw the structural formula of the monomers from which Nylon 6 could be made. State the type of polymerisation involved.

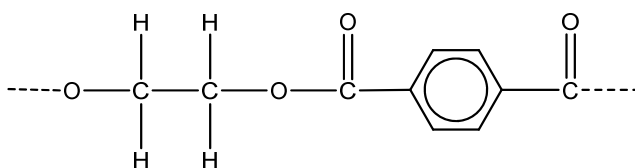


Type of polymerisation: [Condensation polymerisation](#)

[2]

- (b) Draw a displayed formula to show the simplest repeat unit of Terylene.

[1]



- (c) Explain why nylon is prone to creasing while Terylene is wrinkle free.

.....

..... [2]
 Nylon: Polyamide chains are bonded to one another via hydrogen bonding. The hydrogen bonds are broken / disturbed when water penetrate the areas between the chains. New hydrogen bonds are formed while it dries up and form creases.

Terylene: a polyester which do not form hydrogen bonding between the chains, thus is less prone to crease.

Poly(propene) is also a polymer like the synthetic fibre; it is lightweight, durable and widely used in industries for making home articles such as container, buckets and crates.

- (d) Explain why dilute sodium hydroxide will cause holes to appear in containers made from polymers such as Terylene but a poly(propene) container can be used to store sodium hydroxide. [1]

..... [1]
 Terylene (Polyesters) are susceptible to/undergoes hydrolysis in presence of alkali but poly(propene) is unreactive towards alkali.

- (e) State the reason why the disposal of poly(propene) containers in landfill sites is a problem. [1]

..... [1]
 PP is not easily decomposed in nature/break down by microorganisms/non-biodegradable.
 Thus, takes up land space.

- (f) The following is an extract from an article “The safety of nanoparticles in sunscreens: An update for general practice” published on The Royal Australian College of General Practitioners 2016, June 2016.

“Recent media coverage has raised public awareness regarding the safety of sunscreens containing zinc (ZnO) and titanium (TiO₂) nanoparticles. The inorganic filters ZnO and TiO₂ are able to protect against UVA and UVB. As conventional micronised particles in sunscreen, they have several unfavourable characteristics, including difficulty in application, leaving white residue on the skin. This led to the development of smaller nanoparticles, the use of which vastly improves aesthetic and ease of application of sunscreens while maintaining photo-protective qualities.”

- (i) Define nanoparticles.

..... [1]
 Material with all dimensions on 1-100 nm.

- (ii) Explain in your own words why the surface-to-volume ratios makes nanoscale sunscreen attractive.

..... [2]

As surface area to volume ratio is high, a greater amount of a blocking agent (sunscreen) is in contact with the nanoparticle. The denser packing covers the skin more evenly/ does not leave bulky residue behind. (or words to the same effect)

- (iii) Explain why nanoparticles of TiO_2 in sunscreen could present a risk to human health.

..... [1]

it can easily pass through/absorb into pores/ skin/ cell / membranes / veins/ capillaries/ into blood stream AND get into blood and cause cancer / toxic to cells

- (iv) Sunscreen containing nanoparticles should be tested further.

Suggest one reason why some companies that make sunscreens might not want to do more tests.

.....

..... [1]

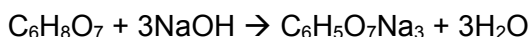
1 of the possible answers:

- Testing is time-consuming
- Testing is expensive
- Testing may result in undesirable outcome / reduce sales (owtte)
- Accept other logical answers

- 3 Citric acid, $\text{C}_6\text{H}_8\text{O}_7$, is a weak tribasic organic acid. It is used as an additive and preservative in various food and drinks to enhance flavour and taste. It provides a fruity tartness to complement fruit flavours and it is also used to mask the unpleasant taste of pharmaceutical products.

However, a high content of citric acid in foods and drinks could damage your teeth. Chemical erosion of the teeth occurs either by the hydrogen ions derived from citric acid or by the citric acid anions which can bind or complex calcium. The acceptable limit for the percentage composition by mass of citric acid in food is 2.5% as this is around the percentage composition commonly found in natural fruits.

A student wanted to find the out the mass of citric acid in a mint candy by doing a titration with sodium hydroxide solution. The equation for the titration is:



He weighed out 5.00 g of the mint candy and crushed the candy into powder. He then dissolved the powder in 100 cm^3 of distilled water, stirred the mixture and filtered it forming solution **A**.

- (a) The student then prepared a standard solution of sodium hydroxide by dissolving 0.104 g of sodium hydroxide pellets in 25.0 cm³ of distilled water. He then made the volume up to 250 cm³ forming solution **B**.

Calculate the concentration of sodium hydroxide in solution **B**.

$$\text{Amount of NaOH in 25.0 cm}^3 \text{ of distilled water} = 0.104/40.0 = 0.002594 \text{ mol}$$

$$[\text{NaOH}] \text{ in solution B} = 0.002594 \times (1000/250) = 0.0104 \text{ mol dm}^{-3}$$

[2]

- (b) The student then found that 20.0 cm³ of solution **A** required 10.40 cm³ of solution **B** for a complete reaction.

- (i) Suggest a suitable indicator for the titration.

Phenolphthalein.

Do not accept methyl orange or universal indicator.

[1]

- (ii) Calculate the mass of citric acid in the 20.0 cm³ of solution **A**.

ECF from (a)(i)

$$\begin{aligned} \text{Amount of Citric acid in 20.0 cm}^3 \text{ of A} &= (10.40/1000) \times 0.0104 \times (1/3) \\ &= 3.605 \times 10^{-5} \text{ mol} \end{aligned}$$

$$\text{Molar mass of citric acid} = 192.0 \text{ g mol}^{-1}$$

$$\begin{aligned} \text{Mass of citric acid in 20.0 cm}^3 \text{ of A} &= 3.605 \times 10^{-5} \times 192.0 \\ &= 0.00692 \text{ g} \end{aligned}$$

[2]

- (iii) Given that the average weight of one piece of mint candy is 0.50 g, calculate the mass of citric acid in one piece of mint candy.

$$\text{Mass of citric acid in 100.0 cm}^3 \text{ of A} = 0.00692 \times 5 = 0.0346 \text{ g}$$

$$\begin{aligned} \text{Mass of citric acid in one piece of mint candy} &= (0.0346/5.00) \times 0.5 \\ &= 0.00346 \text{ g} \end{aligned}$$

[2]

- (c) From your answer in (b) (iii), determine whether the percentage composition by mass of citric acid in the mint candy is above or below the acceptable limit.

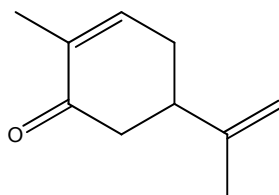
$$\begin{aligned} \text{Percentage composition of citric acid by mass} &= 0.00346/0.5 \times 100\% \\ &= 0.692\% \end{aligned}$$

The percentage composition is **below** the acceptable limit.

[1]

[Total: 10]

4. Carvone is a natural oil produced by the plant *Mentha spicata*. It has the flavour of spearmint and it has important uses in food flavouring.



Carvone

- (a) Carvone reacts with different reagents

- (i) Identify the functional groups present in Carvone.

Alkene and ketone (Do not accept alkane or carbonyl)

[2]

- (ii) Draw structures for the organic products of each compound with the following reagents. State the type of reaction that is happening for each reagent.

Reagent and conditions	Type of reaction	Organic product
Br ₂ liquid, room temperature	Addition	
H ₂ with Ni catalyst, high temperature and high pressure	Reduction	
LiAlH ₄ in dry ether, room temperature	Reduction	

[7]

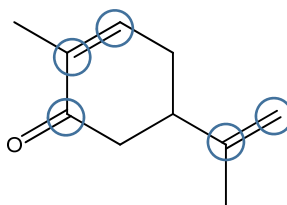
- (b) Suggest why Carvone is highly soluble in hexane.

Carvone has a **bulky hydrocarbon chain/ many C-C and C-H bonds** that enables it to form **instantaneous dipole – induced dipole interactions** with hexane molecules.

[2]

- (c) On the above structure of Carvone, circle one sp^2 hybridised carbon atom.

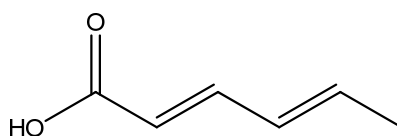
[1]



Carvone

Any one of the 5 circled Cs

- (d) Sorbic acid, or 2,4-hexadienoic acid, is a natural organic compound used as a food preservative.



Sorbic acid

Describe a chemical test that will help you distinguish Carvone from Sorbic acid, including all observations that would be made.

Add **sodium carbonate** to both Carvone and sorbic acid.

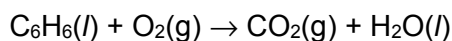
Sorbic acid will give **effervescence**. Gas produced forms a white precipitate with calcium hydroxide.

No observable change will be seen for Carvone.

[2]

[Total: 14]

- 5 (a) Benzene undergoes complete combustion with oxygen as shown by the equation below.



- (i) Explain what is meant by *standard enthalpy change of combustion*.

.....

.....[1]
It is the enthalpy change when one mole of a compound is completely burnt in oxygen under standard conditions of 298K and 1 atm.

- (ii) 1.17 g of benzene was burnt to heat up 250 g of water in a beaker. The initial temperature of the water was 27.6°C. Given that the standard enthalpy change of combustion of benzene is $-3267 \text{ kJ mol}^{-1}$ is and there is 20% heat loss to the surrounding, calculate the highest temperature reached by the water.

[3]

Amount of benzene = $1.17 / (12.0 \times 6 + 1.0 \times 6) = \underline{0.0150 \text{ mol}}$

Total amount of heat evolved = $0.0150 \times (-3267) = \underline{-49.01 \text{ kJ}}$

Amount of heat absorbed by the water = $\underline{0.80 \times 49.005} = 39.20 \text{ kJ}$

Heat absorbed by water = $mc\Delta T$

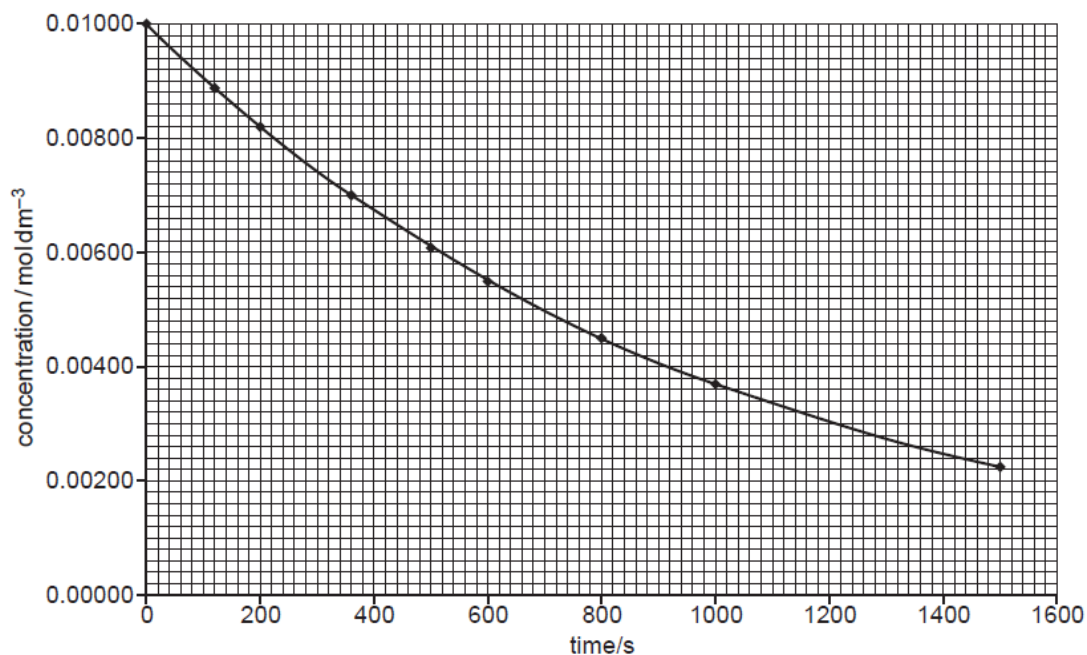
$$39.20 \times 10^3 = 250 \times 4.18 \times \Delta T$$

$$\Delta T = 37.5^\circ\text{C}$$

Highest temperature reached = $\underline{27.6 + 37.5} = 65.1^\circ\text{C}$

- (b) 2-iodo-2-methylpropane undergoes a substitution reaction with hot aqueous sodium hydroxide. Two separate experiments were carried out to study the kinetics of this reaction.

In Experiment 1, sodium hydroxide was used in large excess and the concentration of 2-iodo-2-methylpropane was measured against time. A graph of concentration of 2-iodo-2-methylpropane against time was plotted and shown below.



- (i) Use the half-life method to deduce the order of reaction with respect to 2-iodo-2-methylpropane. Show all your working clearly.

.....

.....[2]

Show clear construction lines to obtain two half lives.

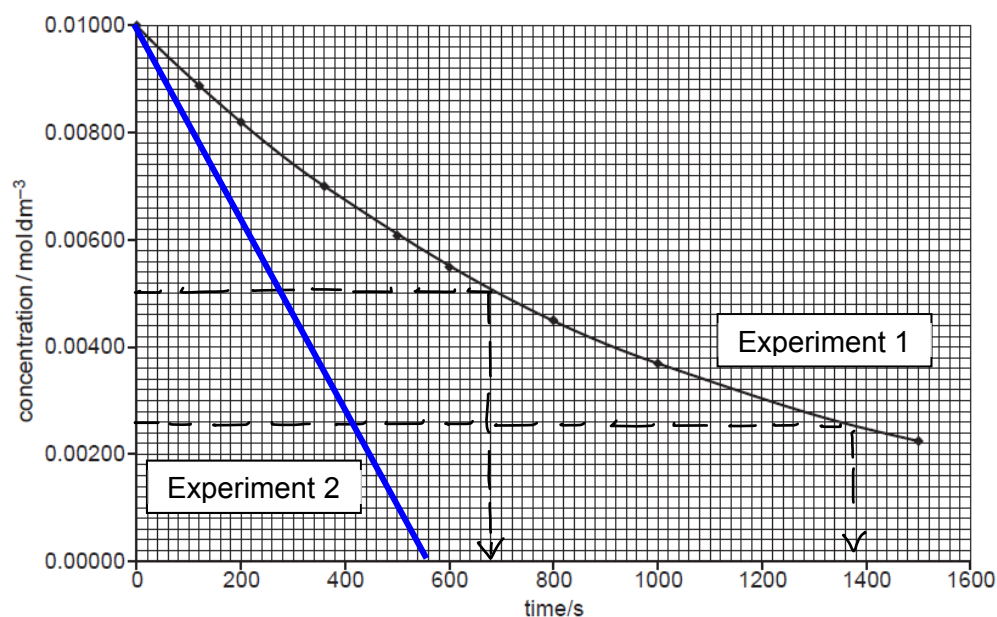
Two approximately constant half-lives obtained $\approx 690 \pm 10 \text{ s}$

Order of reaction w.r.t 2-iodo-2-methylpropane = 1

In Experiment 2, 2-iodo-2-methylpropane was used in large excess and the concentration of sodium hydroxide was measured against time. The following results were obtained.

Time / s	[NaOH] / mol dm ⁻³
0	0.0100
100	0.0082
200	0.0064
380	0.0032
500	0.0010

- (ii) Using the data provided above, plot a graph for Experiment 2 in the graph above. [1]



Points correctly plotted.
Best fit straight line.

- (iii) Using the graph plotted in (b)(ii), determine the order of reaction with respect to sodium hydroxide. Explain your answer.

.....
.....
.....[2]

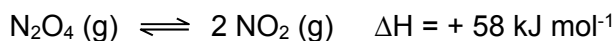
From experiment 2,
[NaOH] vs time graph is a straight line
=> constant gradient => constant rate
(=> rate is independent of [NaOH])

Hence order of reaction w.r.t. NaOH = 0

- (iv) Hence, write an overall rate equation for the substitution reaction.

.....[1]
rate = k [2-iodo-2-methylpropane]

- (c) Colourless N_2O_4 and brown NO_2 are two oxides of nitrogen that can co-exist in equilibrium:



Predict and explain how the colour intensity of the mixture changes, when a reaction chamber containing an equilibrium mixture of N_2O_4 and NO_2 is:

- (i) immersed in boiling water.

.....

.....[2]
 When there is an increase in temperature, equilibrium shifts right to favour the endothermic reaction to absorb the additional heat, producing more NO_2 , leading to an increase in colour intensity.

- (ii) subjected to a sudden compression in volume.

.....

.....[2]
 When there is a sudden compression of the reaction chamber, the pressure of the system increase. By Le Chatelier's Principle, the equilibrium position shifts to the left, favouring the production of fewer gaseous molecules, producing more N_2O_4 , leading to a decrease in colour intensity.

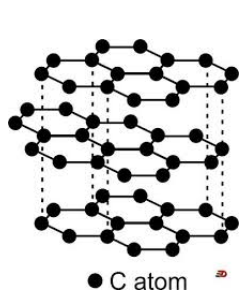
[Total: 14]

Section B

Answer one question from this section, in the spaces provided.

- 5 (a) Describe the structure of graphite. Explain how this structure gives graphite the properties of soft and high electrical conductivity.

You may include labelled diagrams in your answer.



- They have parallel layers of hexagons or shown in diagram
- **Soft:** The adjacent layers are held by **weak id-id interactions** and hence the layers **can slide easily over** one another.
- **Electrical conductivity:** One p electron per C atom (out of the 4 electrons) is **delocalised**, thus is able to conduct electricity well (parallel to layer).

.....

[3]

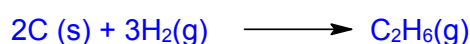
- (ii) Another allotrope of carbon is diamond. State one physical property diamond has in common with graphite and relate the property to the bonding present in their structures.

They have high melting points
 due to the strong extensive covalent bonds between atoms

.....

[2]

- (b) (i) Write an equation, including state symbols, for the reaction with enthalpy change equal to the standard enthalpy of formation for ethane, C₂H₆(g).



.....[1]

- (ii) Suggest in terms of bonding why ethane is a gas at room temperature.

Small amount of energy is required to break the weak instantaneous dipole induced dipole interactions between the ethane molecules.

id-id between molecules AND small amount of energy/weak bonds

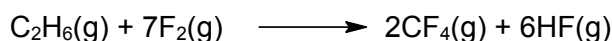
.....
[1]

- (iii) Ethane undergoes a reaction with chlorine. Name the type of reaction that occurs and the conditions used.

Substitution reaction Presence of uv light

.....
[2]

- (b) (i) The enthalpy change for the following reaction is $-2889 \text{ kJ mol}^{-1}$.



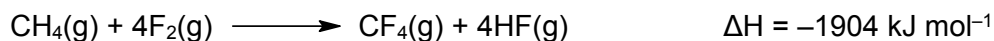
Use this value and the standard enthalpies of formation given to calculate the standard enthalpy of formation of $\text{C}_2\text{H}_6(\text{g})$.

Substance	$\text{CF}_4(\text{g})$	$\text{HF}(\text{g})$
$\Delta H_f^\circ / \text{kJ mol}^{-1}$	-680	-269

$$\begin{aligned} \Delta H_r^\circ &= \sum n\Delta H_f^\circ (\text{products}) - \sum m\Delta H_f^\circ (\text{reactants}) \\ -2889 &= (2 \times -680) + (6 \times -269) - \Delta H_f^\circ \text{C}_2\text{H}_6(\text{g}) && \text{mtd} \\ \Delta H_f^\circ \text{C}_2\text{H}_6(\text{g}) &= -85 \text{ kJ mol}^{-1} && \text{ans} \end{aligned}$$

[2]

- (c) Methane reacts violently with fluorine according to the following equation.



- (i) Using relevant bond energies in the Data Booklet, calculate the bond energy of F-F bond. [2]

Bonds broken (endothermic)	No. of moles	Bond energy / kJ mol^{-1}	Bonds formed (exothermic)	No. of moles	Bond energy / kJ mol^{-1}
C-H	4	4 X 410	C-F	4	4 x 485
F-F	4	4 X F-F	H-F	4	4 x 562

$$\begin{aligned} \Delta H_{\text{reaction}} &= \text{Total bonds broken} - \text{total sum of bonds formed} \\ -1904 &= (4 \times 410) + (8 \times \text{F-F}) - [(4 \times 485) + (4 \times 562)] && \text{mtd [1] / shown in table} \\ \text{F-F bond} &= +161 \text{ kJ mol}^{-1} && \text{ans with correct sign} \end{aligned}$$

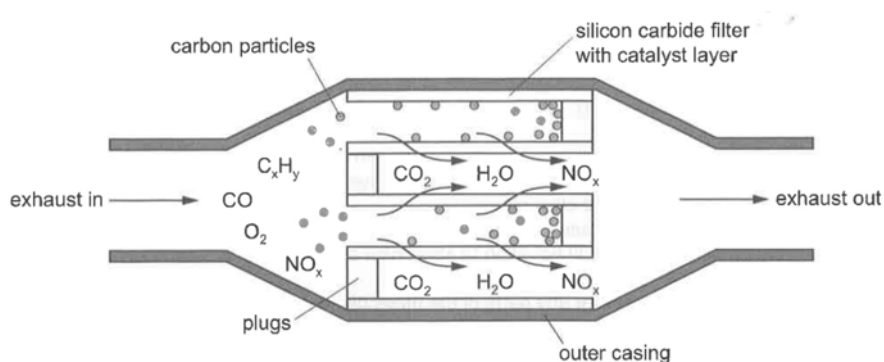
- (ii) A student suggested that actual bond energy of F-F differs from theoretically calculated value in (i). Explain if you agree or disagree with the student.

I agree with the student.

The bond energies used to calculate the (i) are only average values obtained from the Data Booklet.

.....
[1]

- (c) (i) Catalyst are used in car exhaust of modern cars to speed up reaction between polluting gases such as carbon monoxide and dinitrogen oxide, before they reach the end of the exhaust pipe.



- (i) State the type of catalyst involved in this reaction.

..... [1]
Heterogeneous catalyst (catalyst and reactants are in **different phase**)

- (ii) Sketch a Boltzmann distribution curve for the reactants and use it to explain how the use of catalyst speeds up the reaction.

Diagram



Explanation

.....

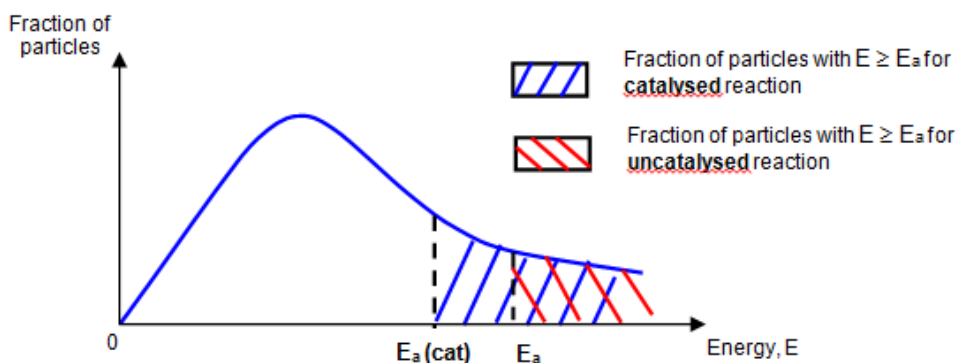
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.....

.....

.....

[4]

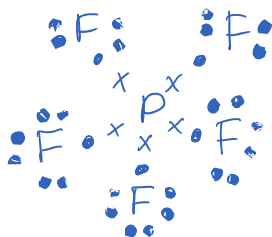


- Catalyst **lowers the activation energy by providing an alternative reaction pathway.** (shown in graph)
 - **Fraction of particles with energy equal to or greater than the activation energy of the catalysed reaction, $E_{a(cat)}$ increases.**
 - **Increased frequency of effective collisions taking place in the reaction**
 - **rate of reaction increases.**
- (ii) Suggest why car manufacturers are required to fit catalytic converters to car exhaust systems.
 Catalytic converters can **remove air pollutants such as CO and NO₂** and convert them into **harmless gases such** as CO₂ and N₂.
- CO will bind with haemoglobin thus preventing the transport of oxygen in humans, or that NO₂ forms photochemical smog or acid rain.

- 6 (a) Phosphorus reacts with fluorine to form PF_3 and PF_5 .

- (i) Draw the dot-and-cross diagram for one molecule of PF_5 . Show only the outer shell electrons.

[1]



- (ii) Explain why the P–F bond is polar.

[1]

F is more electronegative than P / There is a difference in electronegativity between P and F.

- (iii) Predict whether or not a molecule of PF_5 is polar. Explain your answer.

[1]

Non-polar.

The dipole moments cancel out each other.

(specimen paper accepts the molecule is symmetrical)

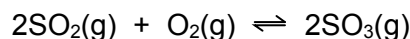
- (iv) Nitrogen reacts with fluorine to form NF_3 only.

Explain why PF_5 exists whereas NF_5 does not exist.

[1]

P is a period 3 element which has energetically accessible vacant 3d orbitals for expansion of octet/ to accommodate more than 8 electrons whereas N is a period 2 element which does not have energetically accessible (3d) orbitals.

- (b) The reaction between sulfur dioxide and oxygen is a reversible reaction and can be described as being in dynamic equilibrium.



A 2-dm^3 reaction vessel containing 0.80 mol of SO_2 , 0.30 mol of O_2 and 1.40 mol of SO_3 is allowed to reach equilibrium at a constant temperature of 1000 K. It was determined that 0.42 mol of SO_2 is present at equilibrium.

- (i) State the meaning of the term *dynamic equilibrium*.

[1]

Dynamic equilibrium is a state of reversible reaction in which the **forward and backward reaction occurs at the same rate** such that the **concentration of reactants and products remain constant**.

- (i) Use the information given to complete Table 6.1.

Table 6.1

	SO ₂	O ₂	SO ₃
Initial amount/ mol			
Equilibrium amount/ mol			

[2]

	SO ₂	O ₂	SO ₃
Initial amount/ mol	0.80	0.30	1.40
Equilibrium amount/ mol	0.43	0.30 – ½ (0.37) = 0.115	1.40 + 0.37 = 1.77

- (ii) Write an expression for the equilibrium constant K_c .

[1]

$$K_c = \frac{[SO_3]^2}{[SO_2]^2 [O_2]}$$

- (iii) Calculate the value of K_c . State the units of K_c .

[2]

$$K_c = \frac{\left(\frac{1.77}{2}\right)^2}{\left(\frac{0.43}{2}\right)^2 \left(\frac{0.115}{2}\right)} = \frac{0.7832}{0.002658} = 294.7 = 295$$

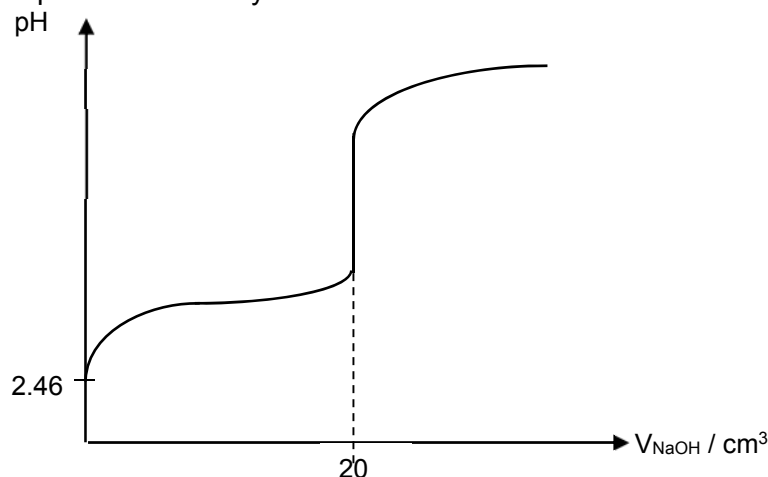
Units for K_c : mol⁻¹ dm³

- (c) (i) State the difference between a weak acid and a strong acid

.....

..... [1]
Weak acid undergoes **partially dissociation** in water to form H⁺ ions whereas strong acid undergoes **complete dissociation** in water to form H⁺ ions.

20.0 cm³ of 0.01 mol dm⁻³ propanoic acid, CH₃CH₂COOH was titrated against 0.01 mol dm⁻³ aqueous sodium hydroxide.



The initial pH of propanoic acid is 2.46.

- (ii) Calculate the initial concentration of hydrogen ions before the titration.

$$\begin{aligned} [\text{H}^+] &= 10^{-2.46} \\ &= 3.47 \times 10^{-3} \text{ mol dm}^{-3} \end{aligned}$$

[1]

During the addition of the first 20 cm³ of sodium hydroxide the mixture is behaving as a buffer.

- (ii) Explain how the shape of the graph show this.

.....

..... [1]
The graph is **relatively flat/ gentle gradient** between 0 to 20 cm³, indicating that **there is little change/increase in pH** during the addition of NaOH.

- (iii) Identify the two species present in the mixture that constitutes to it behaving as a buffer.

..... [1]
CH₃CH₂COOH and CH₃CH₂COO⁻ / CH₃CH₂COONa

- (d) Polyvinyl acetate, PVAc, is a useful adhesive for gluing together articles made from wood, paper or cardboard. The diagram shows a section (not a repeat unit) of the PVAc.

PVAc

- (i) What type of polymerisation made this polymer?

..... [1]
 addition polymerisation

- (ii) PVAc is made from monomer **X**. Draw the structure of the **X**.

[1]

- (iii) PVAc can be converted to polyvinyl alcohol, PVA.

Draw a section of PVA containing at least 2 repeat units and suggest the reagents and conditions needed to make PVA from PVAc in the laboratory.

Section of PVA:



Reagents and conditions: [2]

NaOH(aq), heat or H₂SO₄(aq), heat

- (e) The diagrams below are molecular representations of two types of plastics – thermoplastics and thermosets.

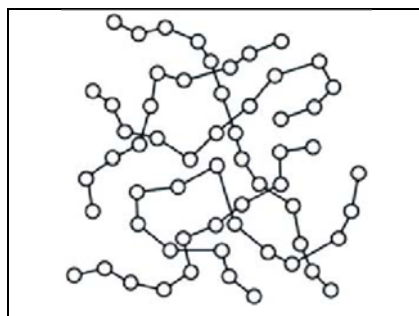


Diagram A

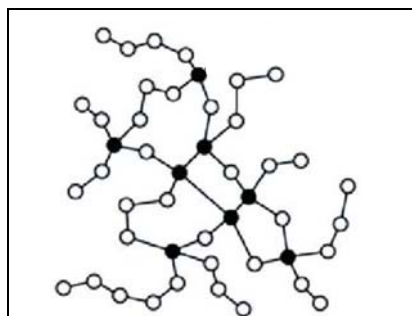


Diagram B

- (i) State and explain which diagram could be the representation for thermosets.

.....

..... [1]
Diagram B. There is **extensive cross-linking** present between the polymer chains.

- (ii) State a difference in physical properties between thermoplastics and thermosets.

.....

..... [1]
Thermoplastics **soften on heating** (and harden on cooling) whereas thermosets **cannot be soften on heating**.

[Total: 20m]